AI-Driven Predictive Maintenance Using Machine Learning

Project Problem Statement

1. Background

Industrial machines may experience unexpected failures that interrupt production, increase maintenance costs, and cause unplanned downtime. Maintenance based only on fixed schedules or after a breakdown may lead to unnecessary servicing or late detection of equipment problems.

2. Problem Statement

The project aims to develop a machine-learning-based predictive maintenance system that uses machine operating parameters to predict whether a failure may occur and, when applicable, classify the possible failure type. The system uses historical machine data to learn patterns associated with failure conditions and produces predictions that can support maintenance planning.

3. Objectives

- Inspect and preprocess the machine predictive-maintenance dataset.
- Use operating features such as air temperature, process temperature, rotational speed, torque, tool wear, and machine type.
- Train an XGBoost model to classify machine records as failure or no failure.
- Train a Random Forest model to classify failure cases into the represented failure categories.
- Evaluate model performance using accuracy, precision, recall, F1-score, confusion matrices, and ROC-AUC where applicable.
- Save trained models and evaluation plots for reuse and project reporting.

4. Proposed Approach

The dataset is inspected and preprocessed, relevant features are prepared, and the data is split into training and test sets. An XGBoost binary classifier predicts failure versus no failure. A Random Forest multiclass classifier predicts a failure category for failure cases. The models are evaluated on held-out test data, and results are summarized using metrics and visualizations.

5. Expected Outcome

The expected outcome is a reusable machine-learning pipeline that demonstrates data-driven machine failure prediction and failure-type classification. The predictions are intended to support maintenance decisions; they do not guarantee that a real machine will or will not fail.

6. Scope and Limitations

The current implementation is based on a historical dataset and trained machine-learning models. Live sensor integration, automatic maintenance alerts, production deployment, and IBM Cloud integration are future enhancements unless separately implemented and verified.

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